Purpose of the DHydROR Algorithm

The Dynamic Hydropower Reservoir Operation Routine (DHydROR) was developed to complement basin-scale hydrological models by explicitly representing hydropower reservoir operations, reservoir sedimentation, and energy generation under predefined operational rules. Conventional reservoir routines in Soil and Water Assessment Tool (SWAT) hydrological model often simplify reservoir behavior through static storage assumptions or predefined release schedules and therefore struggle to realistically simulate large hydropower reservoirs with complex operating conditions. In particular, many existing approaches do not dynamically account for sediment-induced storage loss, spillway operations, turbine-based releases, or reservoir aging, which limits their applicability in sediment-prone and highly regulated river basins.

DHydROR addresses these limitations by simulating the full reservoir water balance, including inflow, storage dynamics, turbine discharge, spillway release, evaporation, seepage loss, and sediment trapping. The algorithm further accounts for progressive reservoir siltation and associated decline in storage capacity, enabling long-term assessment of reservoir functionality and expected useful lifetime. In addition to hydrological processes, DHydROR estimates hydropower generation using turbine discharge and reservoir head conditions, thereby supporting integrated assessment of water, sediment, and energy dynamics within regulated river systems.

DHydROR was designed as a complementary reservoir operation routine for the SWAT model to improve the representation of hydropower reservoirs in watershed-scale simulations. The coupling follows a one-way, offline framework in which SWAT first simulates daily streamflow and sediment inflow at reservoir locations using catchment hydrological processes. These simulated inflows are subsequently transferred to DHydROR, where reservoir operation, water balance, sediment trapping, seepage, evaporation, and hydropower generation are independently simulated using reservoir-specific design information and operational rules.

After reservoir routing is completed, the simulated downstream releases (turbine and spillway outflows) and sediment discharge can be reintroduced into the SWAT river network at the reservoir outlet as modified boundary conditions for downstream routing. This coupling strategy preserves SWAT's strengths in watershed-scale hydrology while extending its capability to represent large hydropower reservoirs with dynamic operation rules and sedimentation-induced storage decline. As a result, the integrated SWAT–DHydROR framework enables more realistic assessment of downstream hydrology, sediment transport, reservoir sustainability, and hydropower generation under existing and future land-use or climate conditions.